

Project Workflow: Weeks 8–10

Phase: Score System, Game Over, Error Handling, Testing & Final Submission

1. Objective

The goal of Weeks 8 through 10 is to finalize the core game mechanics, polish the user experience, ensure system stability through comprehensive testing and error handling, and prepare the project for final delivery. This includes:

- **Week 8:** Implementing a dynamic scoring system based on distance/time and cleared obstacles, developing a dedicated Game Over UI screen with score visualization, and polishing overall visual feedback.
- **Week 9:** Adding robust exception handling for edge cases, performing thorough physical and collision testing, fixing performance bottlenecks, and optimizing frame rendering.
- **Week 10:** Executing full system end-to-end testing across all levels, compiling project documentation and screenshots, and finalizing the submission package.

2. Detailed Task Breakdown

Week 8: Score System & Game Over Screen

- **Score System Engine:** Implemented real-time score tracking using a dual-metric approach combining continuous elapsed time/distance with discrete bonuses awarded for successfully passing obstacle cars or collecting items.
- **Game Over Screen UI:** Created a state-driven Game Over modal overlay that renders automatically upon collision detection, displaying final scores, high scores, level stats, and offering options to restart or return to the main menu.
- **Visual & UX Enhancements:** Added immediate visual feedback mechanisms (e.g., screen flash on impact, animated score counters, and dynamic state transitions) to improve overall player engagement.

Week 9: Error Handling & Testing

- **Exception & State Handling:** Integrated comprehensive try-catch blocks and state-guard logic to handle invalid game states (e.g., unexpected object overlaps, out-of-bounds coordinates, or missing assets) gracefully without crashing.
- **Physics & Collision Validation:** Stress-tested ball physics, momentum, and dynamic collision detection frames across all level configurations to prevent clipping or delayed collision response.

- **Bug Fixes & Performance Optimization:** Cleaned up unused object references, optimized paint rendering loops, and fixed memory leak issues to maintain a stable 60 FPS across long gameplay sessions.

Week 10: Final Testing, Documentation & Submission

- **Comprehensive System Testing:** Conducted end-to-end test runs across all levels (Level 1 to final stage) to verify smooth level transitions, accurate difficulty scaling, correct score tracking, and persistent game states.
- **Documentation & Asset Preparation:** Compiled technical project documentation detailing system architecture, class relationships, level configurations, and user controls, alongside high-resolution UI/gameplay screenshots.
- **Project Delivery:** Cleaned codebase, structured final source files, generated runtime executables (JAR), and completed the project submission package.

3. Deliverables

- **Scoring & Game Over/ Win Module:** Active scoring algorithm and an interactive Game Over/Win UI screen with restart/menu capabilities.
- **Robust & Optimized Build:** A fully tested, crash-resilient game codebase with verified physics, accurate collision bounds, and high frame-rate performance.
- **Final Project Documentation:** Comprehensive documentation file containing project architecture details, feature breakdowns, test logs, screenshots, and final delivery files.

4. Tools & Resources

- **Development Environment:** Java JDK 8+, IntelliJ IDEA / VS Code.
- **Testing & Diagnostics:** Java VisualVM / IDE Profiler for memory management and performance profiling.
- **Documentation & Media:** Markdown/Word editor for project manuals and screen capture software for visual deliverables.